

**Q1. Define a relation T on Z as follows: For all integers x and y
(8 marks)**

$$x T y \leftrightarrow 4 \mid (3x + y)$$

a. Is $1 T 1$?

$$3(1) + 1 = 4 \text{ and } 4 \mid 4$$

Answer : Yes

b. Is $2 T (-2)$?

$$3(2) - 2 = 4 \text{ and } 4 \mid 4$$

Answer : Yes

c. Is $5 T 3$?

$$3(5) + 3 = 18 \text{ and } 4 \nmid 18 \text{ (since } 18 \div 4 = 4.5)$$

Answer : No

d. Is $(-1) T 11$?

$$3(-1) + 11 = 8 \text{ and } 4 \mid 8$$

Answer : Yes

Q2. Let $A = \{0, 1, 2, 3\}$. For each relation below:

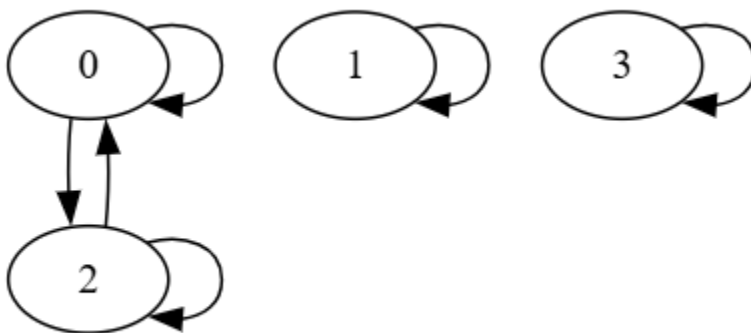
a. Draw the directed graph. ($5 \times 3 = 15$)

b. Determine if it is Reflexive, Symmetric, or Transitive. ($3 \times 3 = 9$)

c. Provide counterexamples for any properties that fail. ($1 \times 3 = 3$)

1. $R_1 = \{(0, 0), (1, 1), (2, 2), (3, 3), (0, 2), (2, 0)\}$

(a) Directed Graph :



(b) Properties :

Reflexive : Yes, because every element has a loop $(a,a) \in R_1$.

Symmetric : Yes. Check: $(0,2) \in R_1$ and $(2,0) \in R_1$; all loops are symmetric

Transitive : Yes. The only non-loop pairs are between 0 and 2.

$(0,2)$ and $(2,0) \rightarrow (0,0)$ in $R1$.

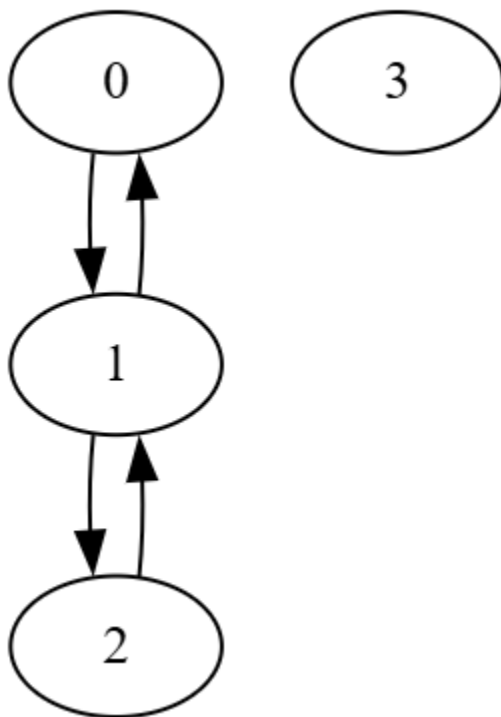
$(2,0)$ and $(0,2) \rightarrow (2,2)$ in $R1$. All other conditions are satisfied trivially.

(c) Counterexamples :

None – all properties hold.

2. $R2 = \{(0, 1), (1, 0), (1, 2), (2, 1)\}$

(a) Directed Graph :



(b) Properties :

Reflexive : No, Counterexample: $(0,0)$ not in $R2$, also $(1,1), (2,2), (3,3)$ missing.

Symmetric : Yes.

$(0,1)$ in $R2$ and $(1,0)$ in $R2$.

$(1,2)$ in $R2$ and $(2,1)$ in $R2$. No other pairs to check.

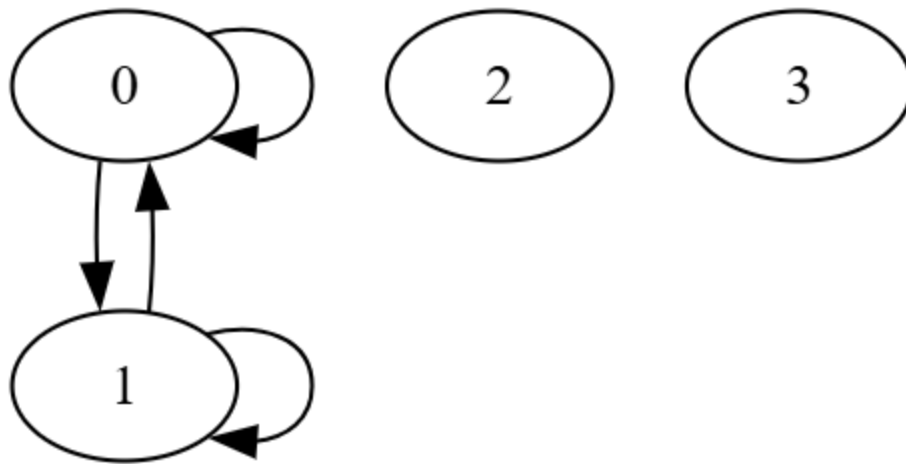
Transitive : No. Counterexample: $(0,1)$ in $R2$ and $(1,2)$ in $R2$, but $(0,2)$ not in $R2$.

(c) Counterexamples :

Only reflexivity and transitivity fails; for symmetry no counterexamples exist.

3. $R3 = \{(0, 0), (1, 1), (0, 1), (1, 0)\}$

(a) Directed Graph :



(b) Properties :

Reflexive : No, Counterexample: $(2,2)$ not in $R3$, also $(3,3)$ missing.

Symmetric : Yes. Check: $(0,1) \in R3$ and $(1,0) \in R3$; all loops are symmetric

Transitive : Yes. Check all possible triples:

$(0,1)$ and $(1,0) \rightarrow (0,0)$ in $R3$.

$(1,0)$ and $(0,1) \rightarrow (1,1)$ in $R3$.

$(0,1)$ and $(1,1) \rightarrow (0,1)$ in $R3$.

Pairs involving 2 or 3 do not create any condition because no edge starts from 2 or 3.

Hence transitivity holds.

(c) Counterexamples :

Only reflexivity fails; for symmetry and transitivity no counterexamples exist.